Shared AI Intelligence: A Demand-Driven Approach to Cost-Efficient Knowledge Reuse in Organizations

A Research Concept Paper for Adaptive, Secure, and Scalable Shared Knowledge Across Organizational Applications

Software Engineering | Artificial Intelligence | Cloud Computing | Cost Optimization

Research Concept Paper — Preliminary Proposal (Not Yet Experimentally Validated)

Abstract

Organizations increasingly use generative AI systems to answer questions, summarize documents, support learning, and automate knowledge-intensive tasks. However, independently processing the same source material across multiple users and applications can introduce redundant computation, repeated context preparation, additional latency, and unnecessary operating costs.

This proposal investigates a shared AI intelligence architecture in which reusable knowledge artifacts are maintained across authorized users, applications, and AI agents within an organization. The proposed approach explores demand-driven knowledge construction: user requests trigger additional processing when needed, while useful results may be retained and reused by subsequent requests. The system makes adaptive decisions about preparation, reuse, updating, and recomputation based on expected demand, processing costs, information freshness, answer quality, and access permissions.

A university is proposed as an initial evaluation environment. The research will compare upfront preparation, independent on-demand processing, fixed reuse strategies, and adaptive demand-driven reuse. The central objective is to determine whether adaptive shared knowledge can reduce total cost per correctly answered query while preserving response quality, freshness, and authorization guarantees.

Keywords: RAG, Shared Knowledge, Multi-Agent Systems, Cost Optimization, Organizational AI, Knowledge Reuse, Access Control

1. Introduction

Generative AI applications frequently depend on external information to answer domain-specific questions. Educational institutions may maintain lecture notes, textbooks, recorded lectures, assignments, research documents, and administrative policies. Multiple students, instructors, applications, and AI agents may need overlapping portions of this material.

A conventional architecture may independently retrieve, prepare, and send relevant context to a language model for each request. Research such as RAGCache and CacheBlend demonstrates that computation associated with reused knowledge can sometimes be retained to improve serving performance. Meanwhile, organizational RAG architectures demonstrate the practical importance of shared infrastructure and document-level authorization.

These developments motivate a broader systems question: **How should an organization decide which knowledge to prepare, when to prepare it, which results to reuse, and when the benefits of reuse justify its cost and operational complexity?**

2. Problem Statement

Two common strategies have important limitations:

- **Upfront preparation:** Documents are processed when uploaded or updated. This can reduce future latency but may incur costs for rarely used material.
- **Independent on-demand processing:** Each request triggers the processing needed. Repeated requests may pay for overlapping work, increasing cumulative cost and latency.

The problem is how to **optimize the lifecycle of reusable AI knowledge under uncertain demand, changing information, authorization constraints, and heterogeneous operating costs.**

3. Research Aim and Objectives

Aim

To investigate whether adaptive, demand-driven reuse of shared AI knowledge can improve the economic and operational efficiency of organizational AI applications without compromising answer quality, information freshness, or access control.

Objectives:

1. Identify and classify existing approaches to shared knowledge, RAG caching, and organizational AI infrastructure.
2. Develop a cost model comparing upfront preparation, independent processing, and demand-driven reuse.
3. Define policies for deciding when to create, retain, extend, refresh, or recompute reusable knowledge artifacts.
4. Investigate how shared knowledge can serve multiple authorized users without violating access boundaries.
5. Evaluate the proposed approach against relevant baselines using repeatable workloads.
6. Identify workload conditions under which shared knowledge provides measurable benefits.

4. Conceptual Architecture

The proposed system consists of six logical components:

1. **Organizational Knowledge Sources** — Documents, manuals, policies, videos, databases, and operational data with metadata and access rules.
2. **Event-Driven Processing Pipeline** — Responds to user queries, document changes, system events, or knowledge gaps via a queue for asynchronous tasks.

High-Level Architecture of the Proposed System

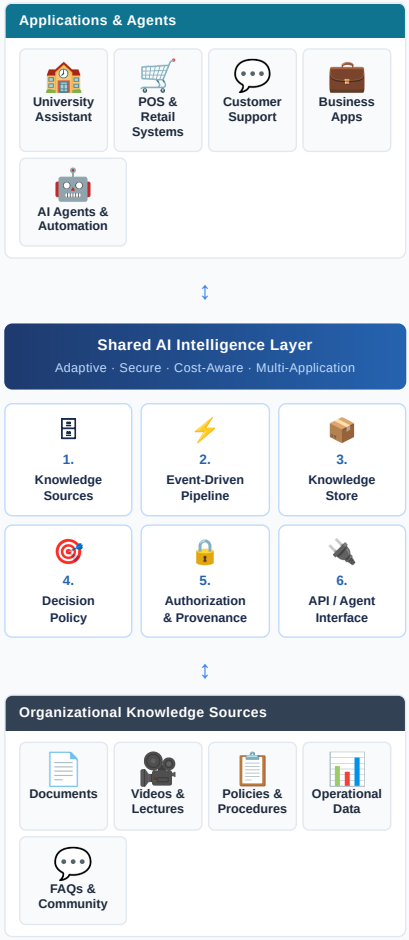


Figure 1. High-Level Architecture of the Proposed System

Core Decision Framework

The adaptive policy evaluates signals to determine the best action for each request:

- **Prepare:** create a reusable artifact when future demand is expected
- **Reuse:** serve from existing artifact when still fresh and authorized
- **Refresh:** update artifact when source material has changed
- **Recompute:** process independently when reuse cost exceeds benefit

Evaluation Scenarios

- **University Knowledge Assistant** — course materials, policies, academic support for students and instructors
- **General Organizational AI** — HR policies, operational manuals, FAQs shared across departments

Expected Contributions

3. **Shared Knowledge and Artifact Store** — Stores embeddings, summaries, verified answers, retrieval paths, and other reusable artifacts.
4. **Adaptive Reuse and Cost-Decision Policy** — Determines whether to reuse, create, refresh, or recompute based on demand, cost, quality, and freshness.
5. **Authorization and Provenance Layer** — Enforces permissions and tracks source, version, and usage history for every artifact.
6. **Application and Agent Interface** — Provides a consistent API for applications, chatbots, and autonomous agents to access shared knowledge.

5. Research Questions

RQ1	**Economic efficiency:** Under what workload conditions does shared, demand-driven knowledge reuse reduce total operating cost compared with upfront preparation and independent on-demand processing?
RQ2	**Adaptive preparation:** Can a policy based on observed demand, expected reuse, and preparation cost make better decisions than a fixed caching or preparation policy?
RQ3	**Quality and freshness:** How do document updates, changing evidence, and artifact expiration affect the quality and reliability of answers generated using shared knowledge?
RQ4	**Organizational security:** How can reusable artifacts be shared across applications and users while respecting document-level permissions and permission revocation?
RQ5	**Organizational scale:** How do the benefits and overheads of shared knowledge change as the number of users, applications, and overlapping requests increases?

6. Economic Model (Preliminary)

Variables

- `C_p` = cost of preparing a reusable artifact
- `C_q` = per-request cost without reuse
- `C_r` = per-request cost with reuse
- `C_m` = maintenance cost (refreshes, invalidation)
- `N` = number of requests benefiting from the artifact

Comparison

C_independent = N × C_q

C_shared = C_p + N × C_r + C_m

Break-even (when C_q > C_r):
N* = (C_p + C_m) / (C_q − C_r)

Shared preparation is cost-effective when: **N(C_q − C_r) > C_p + C_m**. This model is intentionally preliminary and must also account for storage, orchestration, failed processing, concurrency, retrieval overhead, access-control checks, refresh frequency, and differences in answer quality.

7. Research Methodology

Phase 1 Literature Review | Phase 2 Research Refinement | Phase 3 Formal Model | Phase 4 Experimental Design | Phase 5 Evaluation | Phase 6 Reporting

7.1 Baselines

- Upfront preparation
- Independent on-demand processing
- Fixed reuse policy (e.g., TTL caching)
- Adaptive demand-driven reuse (proposed approach)

7.2 Experimental Environment

A university knowledge assistant is a suitable initial case study, with approved course documents and educational materials. Workloads should vary in request frequency, overlap between questions, source size, document update rate, access permissions, and number of concurrent applications.

7.3 Evaluation Metrics

Dimension	Proposed Measurements
Economics	Total processing and serving cost; cost per correctly answered query
Performance	End-to-end latency; time to first token; throughput
Reuse	Artifact reuse rate; cache hit rate; avoided recomputation
Quality	Correctness, evidence support, and retrieval quality
Freshness	Stale-answer rate; time to propagate source updates
Security	Unauthorized retrieval or reuse; access-control violations
Operations	Storage overhead; refresh frequency; failed processing; queue delays

8. Related Work and Research Gap

- **RAGCache** — Studies caching intermediate computation associated with retrieved knowledge to reduce language-model serving latency.
- **CacheBlend** — Studies how precomputed key-value states can be reused across retrieved text chunks while selectively recomputing some states to preserve generation quality.
- **Multi-tenant RAG architectures** — Address shared organizational infrastructure, content isolation, authorization, and operational management across different groups or applications.

1. Comparative economic model for shared knowledge reuse
2. Adaptive decision policy: prepare · reuse · refresh · recompute
3. Evaluation framework: cost + quality + freshness + security
4. Empirical findings on optimal workload conditions
5. Architectural guidance for multi-application knowledge sharing

Work Plan Phases

- **Phase 1** — Literature review: RAG, caching, semantic caching, multi-tenant auth
- **Phase 2** — Research refinement: narrow research question based on literature
- **Phase 3** — Formal model: costs, workload assumptions, decision variables
- **Phase 4** — Experimental design: datasets, baselines, artifact types
- **Phase 5** — Evaluation: implement strategies, collect measurements
- **Phase 6** — Reporting: analyze, document limitations, assess novelty

Adaptive Policy Signals

- Request frequency & overlap between queries
- Preparation cost vs. expected future reuse
- Artifact freshness & source update rate
- Storage and maintenance overhead
- Authorization rules & permission changes
- Answer quality relative to on-demand processing

The potential research gap is the **joint decision problem**: whether a unified policy can make cost-aware decisions across different types of reusable knowledge artifacts while accounting for observed demand, source changes, authorization boundaries, and answer quality.

9. Expected Contributions

1. **A comparative economic model** describing when shared knowledge reuse is beneficial under different workload and cost conditions.
2. **An adaptive decision policy** for selecting between preparation, reuse, refresh, and recomputation.
3. **An evaluation framework** that measures economic efficiency together with quality, freshness, and security.
4. **Empirical findings** identifying workload conditions under which shared organizational knowledge outperforms alternative strategies.
5. **Architectural guidance** for applications and AI agents that share organizational knowledge through controlled interfaces.

10. Scope and Limitations

- Initial focus on organizational document intelligence and a limited set of reusable artifacts.
- A university will serve as the primary use case; principles may apply to other knowledge-intensive organizations.
- Not attempting to develop a new foundation model, solve every form of multi-agent coordination, or transform every possible media type.
- Limitations include workload representativeness, difficulty estimating future demand, variation in model prices, and quality measurement challenges.

11. Conclusion

Shared AI intelligence offers a promising way to investigate the cost and operational consequences of repeatedly processing overlapping information across users and applications. The proposed research focuses on a specific question: how should a shared knowledge system decide when to invest in reusable knowledge, when to reuse it, and when to refresh or recompute it as demand, source material, and permissions change?

A rigorous comparison of upfront preparation, independent processing, fixed reuse, and adaptive demand-driven reuse could establish the conditions under which such a system provides measurable benefits. The work's academic value will depend on verified novelty, a sound experimental design, and evidence that efficiency gains do not compromise quality, freshness, or security.

The immediate next step is to validate the research gap through a focused literature review before committing to an implementation or making claims of novelty.

Preliminary References

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These references provide an initial starting point, not an exhaustive literature review. Before academic submission, bibliographic details, publication status, and additional closely related work should be verified. The AWS article is an industry reference, not a peer-reviewed paper.

This is a research concept paper and not a completed study. The ideas, methods, and results are subject to further investigation, validation, and revision.